UNIVERSITY OF MORATUWA

Faculty of Engineering

Registered Module No: CS4203

Title

Predicting Atrial Fibrillation(AFib) at Edge

Date of Submission:

February 17, 2026

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1 Abstract

Atrial Fibrillation (AF) is the most common sustained cardiac arrhythmia, associated with increased risk of stroke, heart failure, and reduced quality of life. Early prediction of AF episodes, rather than post-onset detection, is crucial for timely intervention and effective management. Existing solutions, however, are primarily cloud-based, raising challenges of latency, data privacy, and limited accessibility, while current wearable devices focus largely on detection rather than prediction. This proposal aims to design and implement an edge-based predictive framework for AF using photoplethysmography (PPG) signals from wearable devices. By developing a hierarchical, resource-aware machine learning pipeline, the project seeks to balance accuracy, efficiency, and interpretability under the computational constraints of edge devices. The proposed methodology spans data acquisition and preprocessing, feature engineering, model training with optimized architectures, and real-time deployment on wearable hardware.

2 Introduction

This chapter presents the background of the study, overview of current approaches andlimitations of current approaches and research gaps.

2.1 Background

In this section we establish the foundational knowledge for this research, covering key concepts such as Human Heart, Atrial Fibrillation, and Continous Monitoring.

2.1.1 The Human Heart

The human heart operates as a pump that circulates blood throughout the body [1]. This complex process is controlled by an electrical system residing within the cardiac muscle itself, allowing the heart to beat even without nervous connections [2]. For each heartbeat, electrical signals are initiated and spread in an orderly manner, causing different parts of the heart to relax and contract, thereby controlling blood flow. The electrical event, known as the wave of depolarisation, triggers muscular contraction.

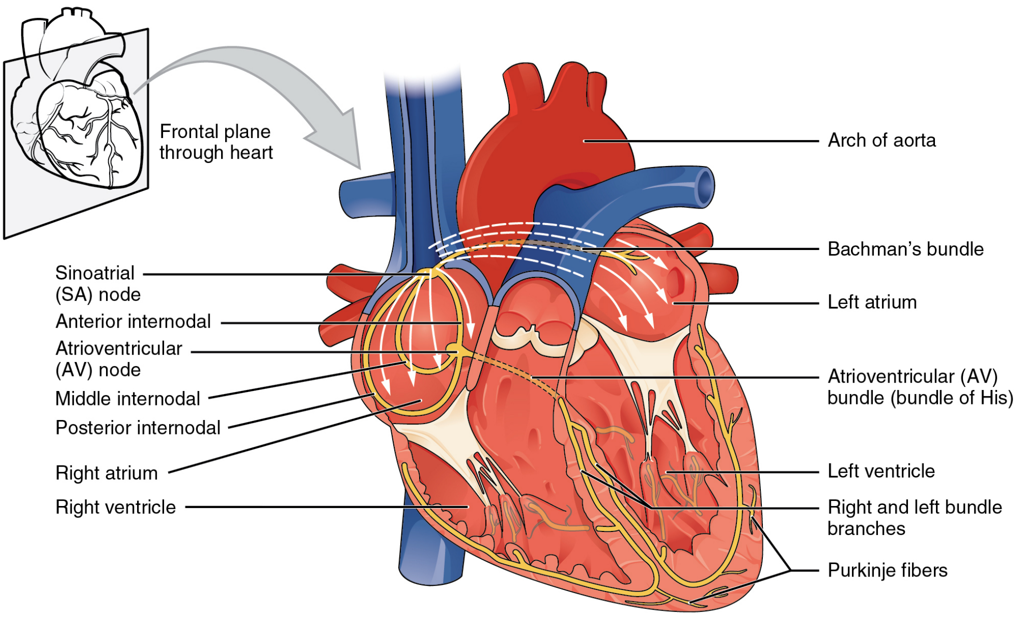

Figure 1: Parts of a Heart [3]

The contraction of the atria begins with the sinoatrial (SA) node, located in the superior and posterior walls of the right atrium [3]. The SA node acts as the heart's natural pacemaker, establishing the sinus rhythm, the normal electrical pattern that leads to heart contractions. After a critical delay, the electrical impulse travels to the atrioventricular (AV) node, found in the inferior portion of the right atrium within the atrioventricular septum. The AV node serves as an important relay point, preventing impulses from directly reaching the ventricles, which allows the atria to finish pumping blood into the ventricles.

From the AV node, the impulse moves to the atrioventricular (AV) bundle, also known as the Bundle of His. This bundle divides into two main branches that provide electrical signals to the left and right ventricles. Finally, the impulse travels through the Purkinje fibers, which rapidly distribute the electrical signal throughout the ventricular myocardium. This quick conduction ensures that all ventricular muscle cells receive the impulse simultaneously, enabling a coordinated contraction that effectively pushes blood into the aorta and pulmonary trunk. Overall, this electrical conduction system is designed to maintain a stable heart rate, which can increase or decrease as necessary to meet the body's oxygen and blood needs.

2.1.2 Heart Electrical Activity

The electrical activity of the heart can be recorded as a complex, compound electrical signal called an electrocardiogram (ECG). This tracing of the electrical signal is obtained by carefully placing surface electrodes on the body, which record the voltage differences generated by the heart. The ECG equipment amplifies these voltages, providing a plot of voltage as a function of time.

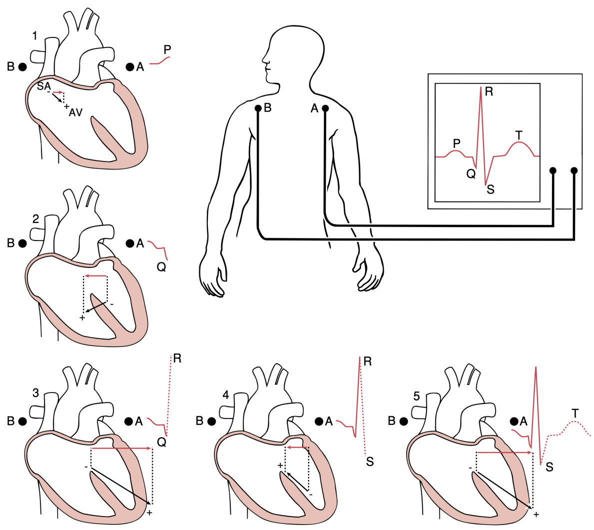

Figure 2: The sequence of major dipoles giving rise to ECG waveforms [2]

The ECG captures dipoles from the heart's electrical activity. Key components of a normal ECG tracing include,

- P wave : Represents atrial depolarization, originating from the SA node, leading to atrial contraction about 25 milliseconds after the wave begins.

- PR segment : The iso-electric period between the P wave and the QRS complex, where depolarization moves through the AV node and bundle system. The PR interval indicates AV conduction time.

- QRS complex : Represents ventricular depolarization, characterised by the initial Q wave (inter-ventricular septum), followed by the R wave (ventricular walls), and concluding with the S wave. Ventricular contraction begins at the peak of the R wave.

- ST segment : The period from the end of the S wave to the start of the T wave, typically iso-electric, indicating all ventricular muscle is depolarized

- T wave : Shows ventricular repolarization, which is less orderly than depolarization. It has a longer duration than the QRS complex because repolarization occurs asynchronously.

2.1.3 Photoplethysmography (PPG) and PPG Signal Characteristics

Photoplethysmography (PPG) is a non-invasive optical technique used to measure blood volume changes in the microvascular bed of the skin [4]. It is based on the absorption, scattering, and transmission of light through biological tissues. A PPG signal is obtained by illuminating the skin with a light source, typically an LED, and detecting the transmitted or reflected light using a photodetector. The resulting waveform reflects variations in blood volume with each cardiac cycle and can be described using the Beer–Lambert law [5].

PPG can be measured in two primary modes: transmissive, where the detector is placed opposite the light source (commonly at fingers, toes, or earlobes), and reflective, where the detector is adjacent to the emitter (suitable for sites like the wrist, forehead, or carotid artery) [6]. The waveform is composed of two main components: a pulsatile alternating current (AC) component, which corresponds to arterial blood volume changes synchronized with the cardiac cycle, and a non-pulsatile direct current (DC) component, which depends on tissue composition, venous blood, and other baseline factors [7].

The morphology of the PPG waveform is highly informative. A typical cycle consists of a rising systolic phase, corresponding to ventricular contraction and the rapid increase in arterial blood volume, followed by a diastolic phase, representing the gradual decrease in blood volume. Key feature points, such as the systolic peak, dicrotic notch, and diastolic peak, provide important insights into vascular compliance, arterial stiffness, and cardiac function [6]. Furthermore, intervals between successive diastolic peaks correspond to pulse-to-pulse intervals, which are closely correlated with the R–R intervals of the ECG, thereby linking mechanical activity (blood volume changes) with electrical activity of the heart.

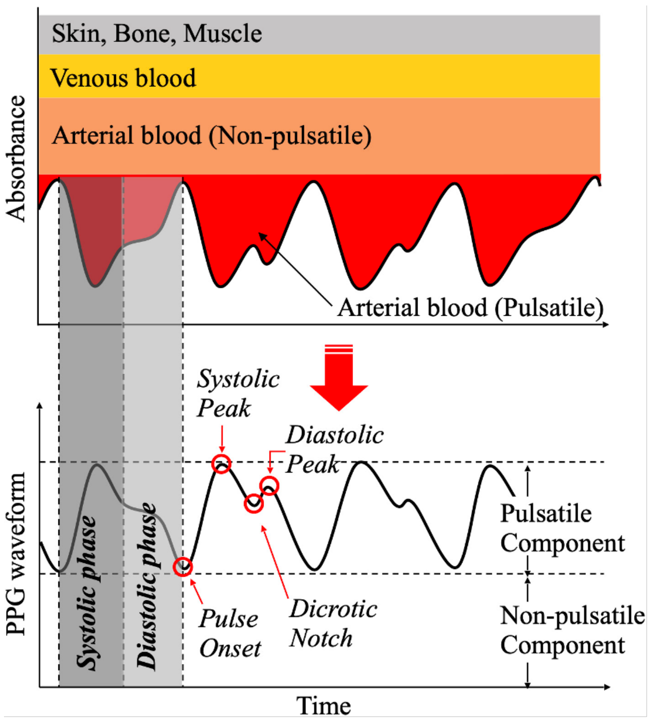

Figure 3: PPG Waveform and Key Feature Points [6]

Key characteristic points of the PPG waveform include:

- **Pulse onset** : The point where pulsation begins, representing the lowest blood volume level before systolic rise.

- **Systolic peak** : The maximum point of the waveform, corresponding to peak arterial blood volume during ventricular systole.

- **Dicrotic notch** : A small inflection on the descending limb, associated with transient backflow before aortic valve closure.

- **Diastolic peak** : A secondary peak occurring after the dicrotic notch, reflecting vascular elasticity and wave reflection.

- **Pulse-to-pulse interval** : The time difference between two successive characteristic points (e.g., diastolic peaks or systolic peaks). This interval corresponds to one cardiac cycle and is strongly correlated with the ECG R–R interval.

2.1.4 Deviations from Normal Electrical Rhythm

An arrhythmia is a deviation from the normal pattern of impulse conduction and contraction of the heart [2]. Normally, the heart beats in a sinus rhythm, which is established by the sinoatrial (SA) node, the heart's natural pacemaker. If an area of the heart other than the SA node initiates an impulse, it is known as an ectopic focus or ectopic pacemaker.

Common types of arrhythmias are Atrial Fibrillation (AF or AFib), Atrial Tachycardia(AT), Tachycardia, Bradycardia, Premature Ventricular Contractions (PVC) and Ventricular Tachycardia (VT).In addition to the these types, there are many other arrhythmias, some of which remain unidentified due to their variability.

2.1.5 Atrial Fibrillation

Atrial Fibrillation (AFib) is the most common type of arrhythmia [8] and is characterized by chaotic, rapid, and irregular atrial rhythms. In AFib, instead of a single, coordinated electrical impulse from the sinoatrial (SA) node, the atria are stimulated by multiple chaotic waves of depolarization (Fig 4). This results in uncoordinated activation of the atria and ineffective contractions [9]. On an electrocardiogram (ECG), AFib is characterized by irregular R-R intervals, the absence of distinct P waves, and irregular atrial activity, which is referred to as fibrillatory waves.

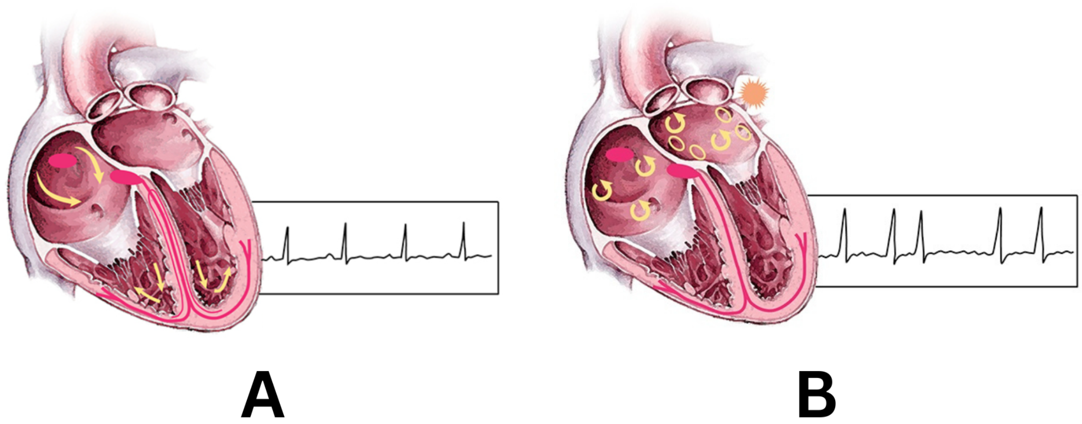

Figure 4: Normal Sinus Rhythm (A) vs Atrial Fibrillation (B) [1]

In addition, AFib is the increasingly prevalent arrhythmia with significant health and socioeconomic impact. It affects approximately 33 million people worldwide and more than 3 million in the United States [10]. The estimated global prevalence was 50 million in 2020, and its incidence and prevalence increased in the United States and globally. The increasing burden is attributed to factors such as the aging population, rising obesity rates, increased detection, and improved survival rates for those with AFib and other cardiovascular diseases.

ECG-derived variables are important markers for AFib. These include PR interval, P-wave duration, area and terminal force [10] .Deviations from the normal ECG pattern, such as the abnormal pattern prior to the QRS complex and increased frequency between QRS complexes, are characteristic of AFib .Other physiological markers like Atrial size and function,Biomarkers (N-terminal fragment B-type natriuretic peptide and high sensitivity C-reactive protein),Atrial fibrosis and Atrial ectopy are included.

AFib has a wide range of adverse effects on heart function and overall health. One of the primary concerns is impaired pumping efficiency, as the chaotic atrial activity inhibits proper atrial contraction, leading to rapid and irregular ventricular rates that compromise the heart's ability to pump effectively. Additionally, AFib significantly increases the risk of thromboembolic events such as stroke and bleeding, particularly due to the formation of thrombi in the left atrial appendage [10].

If left untreated, AFib often progresses from a paroxysmal form to a persistent form, resulting in a higher burden of arrhythmia and poorer outcomes. Clinically, AFib is a common reason for hospitalization, particularly when it occurs suddenly during an illness or after surgery [9]. This can lead to prolonged recovery times and an increased burden on healthcare resources. Overall, AFib is a complex disorder that not only impairs heart function but also significantly contributes to systemic complications, reduces quality of life, and increases the risk of death.

Currently research tasks in this field mainly fall into two categories. The first is detection, which involves identifying patients from all the participants. The second is prediction, which involves in predicting whether AF events will occur in the future.

2.2 Overview of Current Approaches

2.2.1 AFib Detection and Monitoring

Current AFib detection and monitoring approaches involve a combination of traditional clinical interventions and modern technological advancements aimed at early diagnosis and comprehensive management.The initial assessment of patients with AF involves developing treatment strategies with two main goals: preventing thromboembolism and controlling symptoms through either a rhythm control or rate control strategy [9]. Lifestyle and risk factor modification is also a cornerstone of AF management, aiming to prevent its onset, progression, and adverse outcomes. This includes managing obesity, promoting weight loss, encouraging physical activity, smoking cessation, alcohol moderation, and controlling hypertension and other comorbidities.

Traditional Diagnostic Approaches [10],

- **ECG** : AF is diagnosed by electrocardiographic recording, typically showing rapid (over 300 beats/minute), irregular, and often low-amplitude atrial activity with an irregular ventricular response, and no discernible P-waves preceding the QRS complex.

- **Holter Monitors and External Ambulatory Monitors** : Short-term ECG monitoring, followed by continuous ECG monitoring for at least 72 hours, is recommended for screening AF in patients over 65 years of age or those with transient ischemic attack (TIA) or ischemic stroke.

- **Cardiac Rhythm Management Devices** : Pacemakers and defibrillators with atrial leads can detect AF. However, visual confirmation by reviewing intracardiac tracings is necessary to exclude signal artefacts or other arrhythmias, as false positives are possible.

Modern Approaches,

- **Wearables and PPG-based Smart Devices** : Photoplethysmography (PPG) based optical sensors are used to monitor variations in blood volume, allowing the extraction of physiological parameters such as heart rate. PPG is frequently integrated into wearable devices like smartwatches and smartphones, where algorithms can actively and passively detect atrial fibrillation (AF) by analyzing the pulse's temporal and morphological features. However, PPG devices can encounter issues with artifacts and noise, which may lead to false positives and difficulties in distinguishing AF from other arrhythmias.

- **AI and ML Integration** : Artificial Intelligence (AI) and machine learning (ML) techniques are increasingly being utilized to enhance the detection of atrial fibrillation (AF) across various medical settings. These algorithms are applied to a variety of clinical variables, electronic health record (EHR) data, and diagnostic tests.In fact, these algorithms have demonstrated impressive performance in distinguishing AF from other arythmias, show promise for non-invasive and contact-free screening strategies.

Overall, the field of AF detection is rapidly evolving, moving towards more advanced, AI-driven methods that complement traditional approaches to provide earlier, more accurate, and continuous monitoring, thereby improving patient outcomes and management

2.2.2 AFib Prediction

Predicting AFib holds significant importance due to its overall prevalence and significant health and socioeconomic impact, particularly its strong association with cardio-embolic stroke. The focus has shifted towards prediction over mere detection primarily because AF is often subclinical and diagnostically elusive, meaning it can occur without noticeable symptoms or be intermittent, making it challenging to diagnose with traditional methods.

Early warning signs of AFib are crucial for effective management and stroke prevention, as AFib is a major independent risk factor for cardio-embolic strokes. By predicting AF, healthcare providers can implement preventive strategies.Additionally, early prediction allows for timely interventions to reduce the risk of serious complications.By identifying at risk individuals early, healthcare providers can recommend lifestyle changes and manage risk factors.

2.2.3 Current Approaches for Prediction

The current approaches to AF prediction encompass several methodologies aimed at early detection and comprehensive risk stratification. Traditionally, clinical prediction models have been widely utilised to estimate stroke risk and predict AF incidence by integrating various patient characteristics. Building upon these, mathematical and statistical approaches have further refined such risk scores and methods for analysing physiological signals, though issues. More recently, AI and ML based methods have emerged as transformative tools, capable of identifying subtle patterns in vast datasets that are often imperceptible to human observers and traditional statistical analysis. Simultaneously, PPG based prediction approaches, integrated into wearable devices and smartphones, enable real-time, automated AF detection in non-clinical settings.

2.3 Limitations of Current Approaches and Research Gap

2.3.1 Limitations of Centralized Monitoring System

Centralized and cloud-based systems for monitoring AF offer advanced features but face significant challenges. Although high-performance cloud systems can detect AF in as little as 0.02 seconds, latency caused by data transmission to remote servers can delay critical insights needed for early diagnosis. Additionally, the storage of sensitive health data, including from wearables and Electronic Health Records (EHRs), raises privacy and security concerns. Ensuring robust data-sharing practices with strong privacy protections is essential for maintaining patient trust. Connectivity issues further complicate matters, as reliable internet access is necessary for continuous monitoring and data transmission. This reliance can hinder the scalability and accessibility of AF detection tools in areas with limited connectivity or for underserved populations. Developing solutions that function effectively outside traditional clinical settings is crucial to overcoming these barriers.

2.3.2 Prediction at Edge

Edge devices, such as smartwatches, fitness trackers, and smartphones, represent a significant advancement in personal healthcare monitoring, particularly for conditions like AF. These wearable devices incorporate sensors like Photoplethysmography (PPG) to record blood volume variations, and can also integrate single-lead Electrocardiogram ECG capabilities. This enables real-time, automated AF detection in non-clinical settings, significantly expanding screening capabilities beyond traditional clinic walls.

However, a critical characteristic and inherent limitation of these edge devices is their constrained computational power, limited energy, and battery size. These limitations mean that complex algorithms, especially sophisticated deep learning (DL) models, cannot typically be run directly on the devices themselves.

Even though devices like the Huawei Watch Fit 4 Pro incorporate pulse wave arrhythmia analysis, there remains no reliable, FDA-approved method for predicting AF onset fully performed at the edge. Most available systems focus on post-onset detection rather than anticipatory early warning, and they typically rely on cloud processing for complex analysis. This gap highlights both the technical and regulatory challenges in deploying truly autonomous and trustworthy AF prediction algorithms on wearable platforms. Overcoming these challenges will require innovations in model optimization to fit within hardware constraints, algorithms which are robust, low-latency, and privacy-preserving, and rigorous validation to meet stringent approval standards. Until then, most predictive AF models depend on offline processing or hybrid approaches where preliminary data collection occurs on the device but deeper analysis is conducted in the cloud or on external systems.

3 Literature Review

3.1 Machine Learning Approaches for AF Detection

In clinical settings, ML-based algorithms have been used for AF event detection, long-term screening, and identification of subclinical AF episodes on both portable and hospital-grade devices [11]. Risk prediction and patient management are also emerging applications, with models capable of continuously analyzing ECG data and stratifying individuals based on the likelihood of developing AF-related complications. Such approaches outperform traditional techniques, such as 24-hour Holter monitoring or risk stratifying, which often fail to detect likelihood of development of paroxysmal or short-lived AF episodes.

There are numerous approaches that have been used to detect AFib. One of the key approaches is using Machine Learning.. Machine learning methods for atrial fibrillation (AF) detection are generally classified into supervised, unsupervised and semi-supervised learning approaches [11]. Supervised learning has been the most widely used approach, where labeled ECG or PPG data are used to train models such as logistic regression, support vector machines, random forests, and neural networks. These models have shown strong performance in tasks such as AF classification and cardiac risk stratification. In contrast, unsupervised methods, rely on unlabeled data to identify hidden patterns and leverage it for clustering and detect anomalies in large ECG datasets. Semi-supervised learning offers a compromise when labeled data are limited [12].

A Stanford University study trained a deep neural network on more than 90,000 single-lead ECGs and achieved an area under the curve (AUC) above 0.91 across 10 arrhythmia classes [13]. Deep Belief Networks (DBNs), built on a three-layer Restricted Boltzmann Machines, have also shown ability in addressing noise-related challenges in portable ECG devices, achieving sensitivities of 96–100% even in low signal-to-noise environments [14]. In 2021, Mayo Clinic researchers developed CNN-based models trained on over 2.5 million 12-lead ECGs from over 720000 adult patients, which were able to detect AF and other arrhythmias with accuracy outperforming commercial ECG readers and approaching expert over-reading [15].

Overall, reported sensitivities of ML algorithms for AF detection range from 80–100%, with particularly strong results from deep learning methods such as CNNs and DBNs [12] [13] [14]. These performances are consistently comparable to, or in some cases better than, traditional clinical assessment. Importantly, ML algorithms are capable of continuous, scalable monitoring and can filter noise in real-world data, reducing false positives that commonly seen on portable devices. Compared to traditional technologies, which miss up to 85% of AF cases in annual Holter monitoring [11], ML-enabled systems provide a more reliable solution for early detection, long-term monitoring, and clinical decision support.

Metric	**DNN**	**DBN**	**CNN**
Overall Sensitivity	86.1%	73.4–100% (-5 to 10dB) 96.4–100% (-5 to 5dB)	79%
Overall Accuracy	AUC (91%)	75–99.5%	79.4%

Table 1: Performance Comparison of Neural Network Architectures for Atrial Fibrillation Detection [11]

3.2 Wearable Devices for Continuous Health Monitoring

Wearable devices such as smartwatches, rings and wristbands have become increasingly popular tools for continuous health monitoring. Equipped with advanced sensors, they can capture vital physiological parameters such as heart rate, blood pressure, and oxygen saturation [16].

In addition to general wellness tracking, wearable sensor devices provide critical monitoring and diagnostic insights for preventing and managing conditions like arrhythmias, coronary artery disease, and diabetes [16] [17]. They can operate independently or in combination with portable equipment such as smartphones, enhancing accessibility and user convenience [18]. Importantly, wearable devices have emerged as valuable tools for detecting atrial fibrillation (AF), including both symptomatic and asymptomatic cases.

A recent large-scale study highlighted the potential of wearable technologies in healthcare: among users who received notifications of irregular pulses, 34% were subsequently confirmed to have atrial fibrillation (AF) on an electrocardiogram (ECG), and 84% of the notifications corresponded with AF diagnoses [19]. This evidence underscores the clinical significance of wearable devices for the detection and screening of AF.

3.3 Sensing Technologies in Wearables

To understand the clinical use and limitations of wearables, it is important to examine the different sensors that enable physiological monitoring [20]. Accelerometer-based systems, for instance, collect motion data across three axes and use ballistocardiogram (BCG)-based approaches to infer heart rate by measuring repetitive body movements caused by cardiac contractions and blood ejection. Similarly, phonocardiography and seismocardiography (SCG), which record cardiac sounds and vibrations, can aid in identifying structural cardiac abnormalities [19].

3.3.1 PPG in Wearables

Photoplethysmography (PPG) is a commonly employed technique in smartwatches for atrial fibrillation (AF) detection. It operates by illuminating the skin using a light-emitting diode (LED) and capturing the reflected light with a photodetector [21]. The intensity of the reflected light, which varies with blood volume changes in peripheral vessels during the cardiac cycle, forms the PPG signal. This signal represents a pulse pressure waveform, enabling passive, continuous, or semi-continuous heart rhythm monitoring at the wrist. Importantly, the peak of the PPG waveform correlates closely with the R wave of the ECG, thereby allowing estimation of R–R intervals and heart rate variability, which exhibit irregularities in AF episodes [22].

3.3.2 Challenges with PPG in Wearables

Despite the widespread use of PPG, it faces multiple challenges. Poor sensor–skin contact often leads to missing or inconclusive/unclassified signals, while motion artifacts (e.g., from muscular activity) degrade signal quality [23]. PPG accuracy is lower at very high or low heart rates compared to standard ECG-based methods, and ectopic beats can generate irregularities that mimic AF, complicating detection [24].

Additionally, physiological factors such as skin pigmentation [25], aging-related changes (reduced peripheral blood flow, arterial stiffness, altered skin properties), and low signal strength in elderly populations further limit robustness. These factors contribute to poor performance of AF detection and raise concerns about generalizability across populations. [22].

3.3.3 ECG in Wearables

Another major approach to AF detection in consumer wearables relies on recording an ECG. In this method, the back of the smartwatch acts as the positive electrode, while the contralateral fingertip placed on the watch crown serves as the negative electrode, creating a bipolar lead configuration similar to Einthoven's Lead I [16]. The recorded tracing is stored locally and can be analyzed either manually by clinicians or automatically by embedded algorithms. Beyond single-lead ECG, more advanced configurations, such as smartwatch-based wireless six-lead ECGs, have also been explored to improve diagnostic accuracy [22].

3.3.4 Challenges with ECG in Wearables

The small electrode configuration on smartwatches can result in low-amplitude P-waves, making it difficult to classify atrial activity reliably [22]. Moreover, accurate ECG recordings require active user participation, such as placing a finger on the crown, limiting continuous monitoring [16]. Other practical issues include the need for regular device charging and consistent user cooperation to ensure high-quality tracings. **These challenges restrict ECG's practicality for long-term, passive AF screening compared to PPG-based approaches**.

3.4 Utilizing AI and ML for AF Detection at the edge

Artificial intelligence (AI) methods operationalize AF detection on resource-constrained wearables and smartphones by (i) preprocessing PPG/ECG signals, (ii) extracting time– and frequency–domain features or learning representations directly from raw waveforms, and (iii) classifying rhythm using traditional ML (e.g., SVM, random forests) or deep learning (e.g., CNNs/RNNs) [22].

Building on this pipeline, recent work has demonstrated fully edge-deployable AF detection. For example, the *Cardiio Rhythm* algorithm uses a supervised SVM trained on facial and fingertip PPG acquired with an iPhone camera; in a prospective evaluation of 217 cardiology inpatients it achieved 95% sensitivity, 96% specificity, 92% PPV, and 97% NPV for discriminating AF from sinus rhythm, showing that camera-based smartphone PPG can robustly screen for AF without external hardware [26].

Similarly, a smartwatch CNN (SmartRhythm 2.0) trained on heart rate, activity, and single-lead ECG from 7,500 users and validated against insertable cardiac monitors in 24 patients achieved 97.5% episode sensitivity, 97.7% duration sensitivity, and a 39.9% PPV, illustrating that consumer wearables can deliver high-sensitivity AF surveillance directly on the wrist while leaving room to improve precision [27].

3.4.1 PPG based AFib detection using AI and ML

Two landmark pragmatic trials has been highlighted in the literature to show the feasibility of population-scale AF screening using smartwatch-based photoplethysmography (PPG).

The *Apple Heart Study* enrolled over 419000 participants (mean age 41 ± 13 years; 42% women) who wore an Apple Watch. An irregular pulse notification (IPN) required at least 5 out of 6 consecutive irregular 1-minute tachograms within 48 hours. Only 0.52% of the population received an IPN, but this proportion rose to 3.2% among participants $\geq$ 65 years. Those receiving an IPN were mailed an ECG patch for confirmatory monitoring; of these, 34% were diagnosed with AF lasting at least 30 seconds, yielding a positive predictive value (PPV) of 84.0% (95% CI, 76.0–92.0), with a slightly lower PPV of 78.0% in those ≥ 65 years [28].

Similarly, the *Fitbit Heart Study* enrolled 455,699 participants (median age 47 years; 71% women). Its algorithm used more continuous PPG monitoring and stricter criteria, requiring at least 30 minutes of irregular rhythm to trigger notification. One percent of participants (3.6% of those ≥ 65 years) received an IPN, after which they were mailed a confirmatory ECG patch. Among returned patches, 32% were diagnosed with AF, yielding an overall sensitivity of 67.6%, specificity of 98.4%, and PPV of 98.2% (95% CI, 95.5–99.5) [29].

Beyond large-scale pragmatic trials, several smaller validation studies have examined smartwatch performance in higher-risk cohorts under controlled conditions. [30] evaluated the Samsung Galaxy Watch Active 2 in 204 participants with either known AF or elevated risk factors (age ≥ 65, hypertension, diabetes, heart failure, or coronary artery disease). Continuous PPG monitoring demonstrated a sensitivity of 87.8% and specificity of 97.4%. When combined with on-demand single-lead ECG, diagnostic performance improved further, with sensitivity 96.9% and specificity 99.3% against a 28-day ECG patch reference standard.

These studies confirm that consumer wearables using PPG can achieve high specificity and clinically meaningful PPV for AF detection at scale.

3.5 Limitations of AF detection using Smartwatches at the Edge

(i) Clinical perspective. From a clinical standpoint, the absence of consensus guidelines on smartwatch-based AF screening limits physician confidence in using these devices as diagnostic tools [31] [32]. False positives risk overwhelming healthcare providers with unnecessary follow-up visits, while false negatives may provide false reassurance. Therefore, it is crucial to optimize the parameters of smartwatch algorithms to minimize false positives and prevent physician overload [22]. Additionally, inconclusive tracings remain common, often requiring cardiologist review to confirm diagnoses. The long-term clinical impact of smartwatch-detected AF, particularly in asymptomatic patients, is not yet fully established [33].

(ii) Data management. Smartwatch data are not yet fully integrated into electronic health record (EHR) systems. This raises issues of data ownership, compliance with HIPAA regulations, and maintaining patient confidentiality and privacy. Proposals include partnerships between health systems and digital startups, as well as the introduction of "digital health counselors" to guide patients and providers in managing wearable-derived data streams [22].

(iv) Government and regulatory challenges. From a policy perspective, the FDA does not classify smartwatches as medical devices but rather as wellness tools, granting them expedited approval under the Digital Health Software Precertification Program. Furthermore, national guidelines (e.g., ESC vs. USPSTF) remain inconsistent on whether AF screening with wearables is beneficial, leaving uncertainty about reimbursement and adoption at a health system level [22].

Together, these challenges underscore the need for optimization of AI algorithms and the development of robust data infrastructure. Equally important is the interpretability of AI systems, particularly in healthcare settings where providers prefer algorithms that offer transparent insights into their decision-making processes. This aligns with the growing demand for explainable AI in clinical environments and highlights another significant limitation.

3.6 Prediction of AF

3.6.1 Traditional Clinical Risk Scores

Clinical data has historically played a central role in predicting atrial fibrillation (AF). Traditional risk models such as the Framingham Heart Study (FHS) [34], Atherosclerosis Risk in Communities (ARIC) [35], CHARGE-AF [36], C_2HEST [37], and HATCH scores [38] have relied on easily obtainable patient characteristics including age, sex, ethnicity, body mass index, blood pressure, smoking status, comorbidities (e.g., diabetes, coronary artery disease, heart failure), and electrocardiographic parameters such as PR interval [39].

However, their generalizability across diverse populations varies, as seen with C_2HEST, underscoring the need for adaptation or complementary use of machine learning approaches.These models typically achieve area under the receiver operating curve (AUC) values of 0.70–0.78, reflecting moderate discriminative ability [39].However, their generalizability across diverse populations varies, as seen with C_2HEST, underscoring the need for adaptation or complementary use of machine learning approaches.

Table 2: Summary of traditional clinical AF risk scores and their predictive performance [39]

Risk Score	**Variables**	**Original Study (AUC)**	**Validation Study (AUC)**
FHS [40]	Age, sex, BMI, SBP, hypertension treatment, PR interval, murmur, CHF	4,764 US patients, AUC = 0.78 (95% CI: 0.76–0.80)	49,599 US patients, AUC = 0.73 [41]
CHARGE-AF [36]	Age, ethnicity, height, weight, BP, smoking, antihypertensive use, DM, CHF, MI	18,556 US patients, AUC = 0.765 (95% CI: 0.748–0.781)	114,475 Netherlands patients, AUC = 0.74 (95% CI: 0.73–0.74) [42]
C_2HEST [37]	CAD/COPD, hypertension, age, CHF, hyperthyroidism	471,446 Chinese patients, AUC = 0.75 (95% CI: 0.73–0.77)	1,047,330 Danish patients, AUC = 0.59 (95% CI: 0.59–0.59) [43]
HATCH [38]	Hypertension, age, stroke/-TIA, COPD, CHF	670,804 Taiwanese patients, AUC = 0.716 (95% CI: 0.71–0.72)	692,691 Taiwanese patients, AUC = 0.77 [44]

3.6.2 Risk Scores using ML

In contrast, machine learning (ML) approaches leverage the richness of electronic health records (EHRs) to enhance predictive power. For example, Tiwari et al. [45] applied ML to a cohort of nearly 2 million patients, incorporating over 200 clinical features, achieving an AUC of 0.79. Similarly, Sekelj et al. [46] used ML on primary care records from over 2 million patients in the UK, reporting AUCs up to 0.87, outperforming conventional scores. Hill et al. [47] further demonstrated that incorporating time-varying covariates into ML models improved predictive performance (AUC 0.827) compared to the CHARGE-AF score applied to the same population (AUC 0.725).

3.6.3 Cardiac Imaging Approaches for AF Prediction

Studies show that structural abnormalities of the heart detected by cardiac imaging modalities are strongly associated with atrial fibrillation (AF). Echocardiography has been extensively studied, with parameters such as left atrial volume, diastolic dysfunction measures, ventricular wall thickness, and strain imaging shown to predict new-onset AF [48–50]. Non-conventional measures such as total atrial conduction time have also been associated with AF incidence in smaller cohorts [51].

Cardiac CT provides insights into atrial and pulmonary vein morphology. While results are mixed regarding its predictive value for AF recurrence after ablation, measures such as left atrial thickness and low-voltage areas have been linked to progression from paroxysmal to persistent AF and ablation outcomes [52, 53].

Cardiac MRI, with its ability to characterize myocardial tissue, has been particularly valuable in assessing left atrial fibrosis via late gadolinium enhancement. One study showed that fibrosis $>6\%$ was predictive of new-onset AF, achieving an AUC of 0.67, which improved to 0.80 when combined with hypertension history and left ventricular ejection fraction [54].

Recent applications of machine learning (ML) in imaging analysis have further advanced prediction. For instance, ML algorithms applied to cardiac CT morphology in patients undergoing AF ablation achieved AUCs ranging from 0.78 to 0.87 for predicting AF recurrence [55, 56]. These studies highlight the potential of ML to extract subtle imaging biomarkers beyond conventional interpretation.

3.6.4 Limitations of Cardiac Imaging Approaches

However, there have not been investigations in the use of machine learning in cardiac imaging to predict new onset AF [39].

- **Small sample sizes and procedural focus** – Most imaging studies are limited to small cohorts or peri-ablation populations, restricting generalizability.
- **Lack of systematic use in population studies** – Imaging has not yet been integrated into large-scale, prospective AF risk models. Unlike clinical and EHR-based data, imaging is rarely used in population-wide prediction studies.
- **Cost and feasibility constraints** – Routine use of CT and MRI for asymptomatic AF screening is not feasible due to high cost, limited availability, and radiation exposure (in the case of CT).
- **Selection bias** – Imaging studies are typically conducted in patients with existing cardiovascular indications, which may overestimate predictive performance compared to general populations.
- **Data complexity** – Processing imaging data for ML requires specialized pipelines, increasing technical barriers compared to ECG or EHR-based approaches.

3.6.5 Electrophysiological Data Approaches for AF Prediction

Electrophysiological abnormalities are strongly associated with atrial fibrillation (AF). Conventional studies have shown that electrocardiographic (ECG) features such as P-wave duration, dispersion, amplitude, and premature atrial contraction (PAC) morphology and frequency are predictive of new-onset AF, achieving AUCs ranging from 0.69 to 0.87 in different cohorts [57–59]. For example, frequent PACs identified on Holter monitoring have been shown to increase AF risk, although predictive performance in some cohorts remained modest (AUC = 0.58) [60].

Beyond surface ECG, intracardiac electrograms obtained during electrophysiology studies have demonstrated that features such as dominant frequency, regularity index, and organizational index of fibrillatory signals can predict AF recurrence following ablation [61]. However, these approaches are inherently invasive and thus limited to selected patient populations undergoing electrophysiological procedures.

Ongoing Methods Using ML and Deep Learning

Recent research has increasingly leveraged **machine learning (ML) and deep learning** to improve prediction from electrophysiological data:

- **Convolutional Neural Networks (CNNs) on ECG**: A landmark study from the Mayo Clinic analyzed over 600,000 ECGs in sinus rhythm using a CNN, achieving an AUC of 0.87 for new-onset AF prediction, which increased to 0.90 when multiple ECGs per patient were used [62].

- **Heart Rate Variability (HRV) Features + ML Classifiers**: Ebrahimzadeh et al. extracted **linear, non-linear, and time-frequency HRV features** from extended ECG recordings in patients with paroxysmal AF. Their ensemble ML model achieved an accuracy of 98.2%, outperforming conventional classifiers such as SVM, k-NN, and multilayer perceptrons [63].

- **Time-varying covariates in neural networks**: Dynamic ML models that incorporate **temporal patterns in ECG and comorbidities** have also been proposed. By allowing input covariates to vary over time, these models improved predictive accuracy compared to static ECG feature-based approaches [47].

Unlike traditional feature-based analysis, these deep learning models can process raw ECG waveforms directly, uncovering subtle and complex electrophysiological patterns that are not readily apparent to human interpretation.

Recent evidence suggests that integrating multiple data modalities can yield superior predictive performance for atrial fibrillation (AF) compared to single-source approaches. This principle is increasingly being extended to machine learning (ML)-based AF prediction, where algorithms capable of simultaneously incorporating **clinical, electrophysiological, and imaging data** are expected to outperform siloed approaches [39].

However current deep learning models, particularly convolutional neural networks, often function as **"black boxes,"** limiting clinical trust and mechanistic insights. [39] mentions future developments in explainable AI (XAI) may enable these algorithms to highlight key predictive features, facilitating both clinical adoption and new pathophysiological understanding [39].

Previous studies have shown that the prognostic value of ECG-derived markers such as P wave axis variation and P wave dispersion (PWD), which reflect intra- and inter-atrial conduction abnormalities. Increased PWD has been shown to correlate with older age, diabetes, hypertension, left atrial enlargement, ventricular size, and diastolic dysfunction, and is a strong predictor of AF recurrence, particularly in females and patients with higher QT dispersion [64] [65]. Since diastolic dysfunction can also be detected by PPGs, it can be used as a noninvasive marker for early AF prediction.

Overall, utilizing deep learning methods can effectively reduce the workload of manual feature extraction and enhance the model's ability to handle noise interference. However, challenges remain: although these approaches often achieve high accuracy, their recall is typically low, suggesting difficulty in detecting positive AF samples. Moreover, ECG waveforms present diverse morphologies, and AF episodes vary significantly across patients, limiting the generalization ability of existing models. To address these issues [66] has proposed a method combining Recurrence Plots (RP) and ResNet architecture for AF prediction. It has achieved a prediction accuracy of 97% on the PAF Prediction Challenge Database (AFPDB) [67].

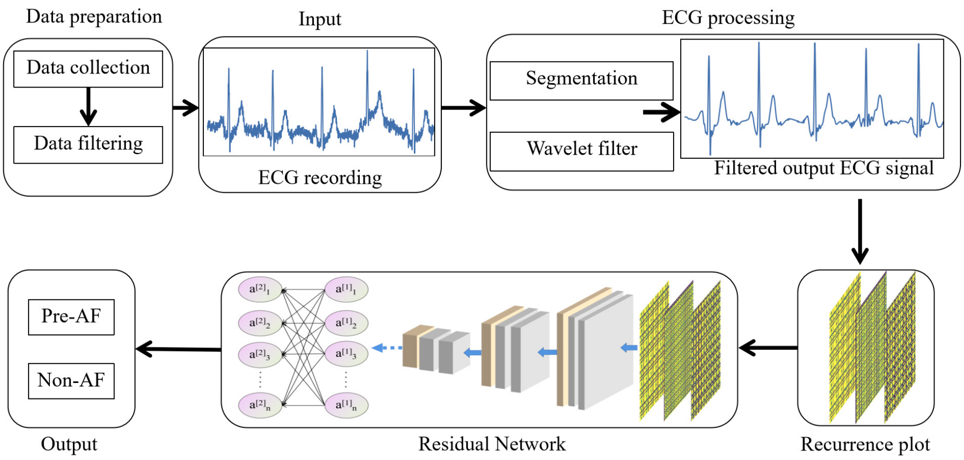

Figure 5: Data processing flowchart of [66]

[66]'s data processing pipeline is illustrated as in Figure 5. Their workflow consists of two main stages: data preparation and data input.

First, ECG signals of varying durations are divided into fixed 10s segments. To minimize inter-database variability, each segment was normalized (Min-max normalization).

$$x_i' = \frac{x_i - \min(x)}{\max(x) - \min(x)} \tag{1}$$

The original ECG signals are often contain noise by baseline drift, powerline interference, and muscle artifacts, which manifest as low-frequency trends or high-frequency fluctuations. To address this, an adaptive wavelet-based denoising approach was applied in their approach. Specifically, the Daubechies (db5) wavelet was used for multi-level signal decompositionas because of its similarity to the QRS complex and concentration of energy in the low-frequency range.

$$T(a,b) = \frac{1}{a} \int_{\mathbb{R}} f(t)\, \varphi\left(\frac{t-b}{a}\right) dt, \tag{2}$$

A soft-thresholding function was then applied to the wavelet coefficients, followed by signal reconstruction to achieve noise removal.

$$T = \delta\sqrt{2 \log N} \tag{3}$$

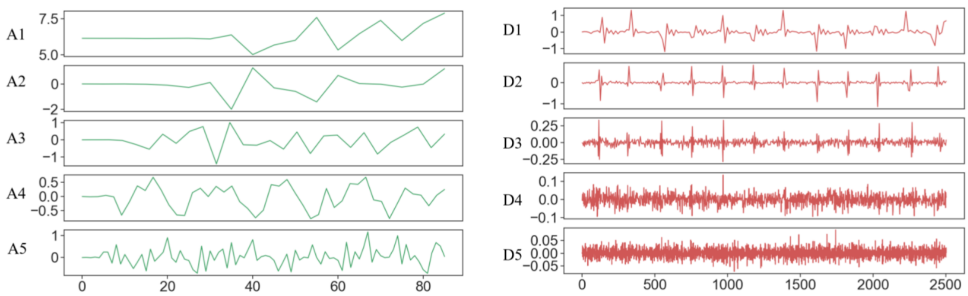

Figure 6: Wavelet decomposition: Low-frequency baseline drift and high-frequency noise are separated into approximation and detail coefficients [66]

After denoising, the 1-D ECG time series was transformed into a 2-D recurrence plot (RP) matrix, which captures the temporal dynamics of the signal in a spatial form.

RP is considered an important technique used to study the periodicity, chaotic nature and non-stationary characteristics of the time series in the literature [68–71].

The key to the RP lies in obtaining the recurrence matrix. For a given time series $x = x(i)$, $1 \leq i \leq N$ with length N, where N represents the length of the time series. The m-dimensional state space representation of the signal is given by Equation (4).

$$u_m(i) = \{x(i), x(i+\tau), \ldots, x(i+(m-1)\tau)\} \tag{4}$$

In Equation (4), $1 \leq i \leq N-(m-1)\tau$, and τ represents the time delay parameter, while m represents the embedding dimension parameter. In this study, τ was set to 3, and m to 3. The distance between any two vectors in the phase space is defined as Equation (5).

$$R_{i,j} = \Theta(\varepsilon - \|u_m(i) - u_m(j)\|) \tag{5}$$

where

$$\Theta(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases} \tag{6}$$

In the given context, $i, j = 1, 2, \ldots, N-(m-1)\tau$, ε represents the predefined threshold, which was chosen as 10% of the maximum distance between any two vectors in the phase space by the authors. $\|\cdot\|$ represents the Euclidean norm, and $\Theta(x)$ is the Heaviside function defined by Equation (6).

By applying the formula, the distance matrix $[N - (m-1)\tau] \times [N - (m-1)\tau]$ is transformed into a binary matrix, ranging from 0 to 1.

Here furthermore, they have obtained a set of quantitative evaluation metrics from the perspectives of point density, diagonal patterns, and horizontal and vertical structures of the recurrence plots.

The Recurrence rate (REC) is the percentage of recurrent points, and is defined as Equation (7), where M is the dimension of the recurrence matrix.

$$\mathrm{REC} = \frac{1}{M^2} \sum_{i,j=1}^{M} R_{i,j} \tag{7}$$

Determinism (DET) measures the percentage of determinism, which quantifies the proportion of recurrence points that form diagonal structures in the recurrence plot; here $l_{\min}$ is 2, and is defined as Equation (8).

$$\mathrm{DET} = \frac{\sum_{l=l_{\min}}^{M} l \cdot p(l)}{\sum_{i,j=1}^{M} R_{i,j}} \tag{8}$$

RP Entropy (ENTR) is used to quantify the degree of disorder or complexity in the RPS, and is defined as Equation (9).

$$\mathrm{ENTR} = -\sum_{l=l_{\min}}^{M} p(l) \ln p(l) \tag{9}$$

Trapping time (TT) refers to the duration or time interval that a system's trajectory remains in close proximity to a certain state before moving away, and is defined as Equation (10).

$$\mathrm{TT} = \frac{\sum_{v=v_{\min}}^{M} v \cdot p(v)}{\sum_{v=v_{\min}}^{M} p(v)} \tag{10}$$

Laminarity (LAM) is diagonal lines or clusters of points that indicate periods during which the system's trajectory remains relatively constant or laminar, and is defined as Equation (11).

$$\mathrm{LAM} = \frac{\sum_{v=v_{\min}}^{M} v \cdot p(v)}{\sum_{i,j=1}^{M} R_{i,j}} \tag{11}$$

L_{mean} measures the average length of the diagonal lines whose lengths exceed the certain threshold $l_{\min}$, and is defined as Equation (12).

$$L_{\text{mean}} = \frac{\sum_{l=l_{\min}}^{M} l \cdot p(l)}{\sum_{l=l_{\min}}^{M} p(l)} \tag{12}$$

These RP images were then used as inputs to the deep learning network for AF prediction. Their deep learning architecture [66] comprises of three main components: CNN layers, improved ResBlocks and fully-connected(FC) layers. The first part of the ResNet architecture was mainly used to process the input data and extract information such as Recurrence rate, entropy, trapping time, and Lmax of the RP. The second part consists of ResBlocks.

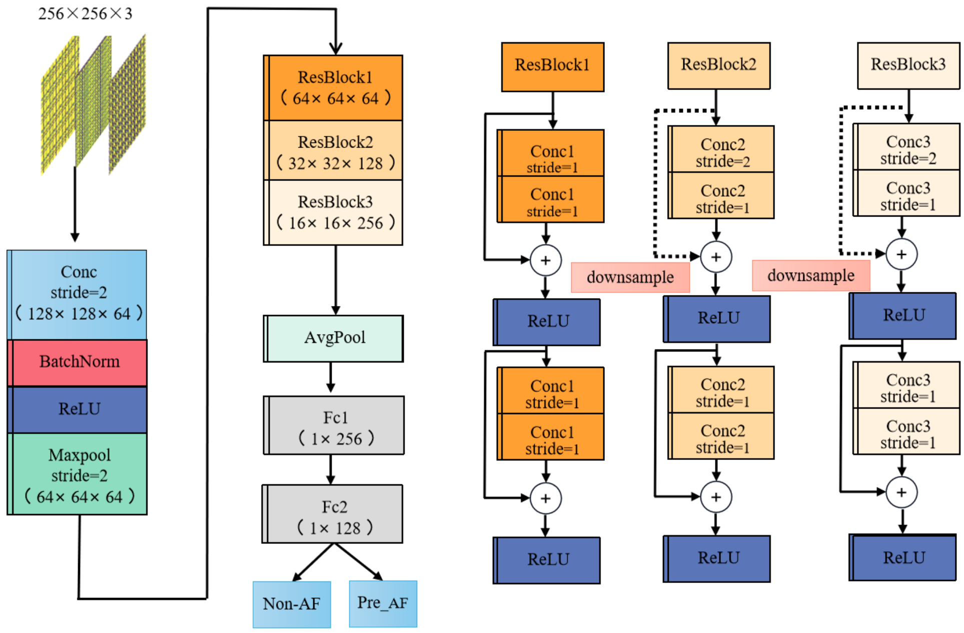

Figure 7: ResNet Architecture for AF prediction [66]

To evaluate the method, they have extracted statistics of each metrics. As in the table, the following characteristics could be identified; The recurrence rate of Pre-AF windows is higher than that of Non-AF windows, The entropy of the corresponding Pre-AF windows is higher due to complexity in RP structure, The trapping time shows a significant range in Pre-AF windows that could be easily used to distinguish AF and Pre-AF windows.

Table 3: Statistical comparison of features between Pre-AF and Non-AF groups [66]

Feature	**Pre-AF (Mean ± STD)**	**Non-AF (Mean ± STD)**	**p-Value**
Recurrence rate	0.544 ± 0.182	0.479 ± 0.127	0.00095
Determinism	0.999 ± 0.033	0.998 ± 0.035	0.814
Entropy	0.462 ± 0.169	0.441 ± 0.141	0.042
Trapping time	0.726 ± 0.028	0.605 ± 0.023	0.00037
Laminarity	0.999 ± 0.016	0.997 ± 0.015	0.712
Lmax	0.369 ± 0.125	0.324 ± 0.129	0.008
Lmean	0.172 ± 0.136	0.166 ± 0.128	0.078

To date, the model described in [66] is only capable of preliminary screening and cannot perform real-time monitoring for patients.

However, this approach relies on the full ECG waveform to generate recurrence plots for prediction of AF. The recent literature include approaches that use the advantages of R-to-R intervals(RRI) [72]in modeling heart rate dynamics to generate Recurrence Plots and predict AF. RRI signals can be acquired from wearable and affordable devices, such as smartwatches or fitness bands, enabling continuous, real-time monitoring and early warning for patients in daily life.

One of this approach [73] propose as an automated deep learning–based Convolutional Neural Network(CNN) system named WARN designed to predict AF onset from RRI signals. The model processes 30-s RRI segments in every 15 s, and outputs a probability representing the imminent risk of AF. A key characteristic of this procedure is that it can capture a continuous rise in risk probability well before AF onset, issuing an early warning once the score crossed a predefined threshold. On a test set of 70 patients and two external validation sets comprising 33 patients, WARN successfully predicted AF onset on average 31 and 33 minutes in advance, achieving accuracies of 83% and 73%, respectively.

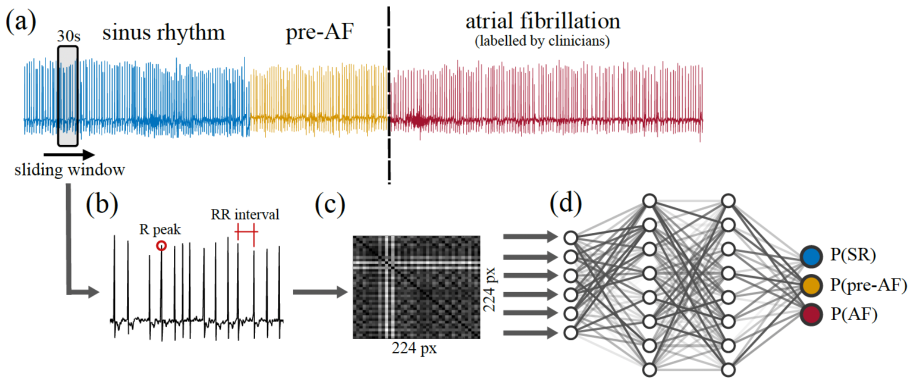

Figure 8: Pipeline of the first stage of WARN. (a) ECG recordings are segmented into three classes: SR, pre-AF, and AF. (b) R peaks are automatically detected within sliding windows of the ECG, enabling conversion to RRI data. (c) Recurrence plots are generated from each RRI window. (d) A deep CNN is trained on these recurrence plots, producing probabilities for each class (SR, pre-AF, and AF) for the sampled data. [73]

WARN method consists of a pipeline which is subdivided into four stages. First ECG lead-II recordings with a sampling rate of 128Hz and a resolution of 10-bits we acquired. Then they were pre-processed with 0.5-40Hz band-pass filter to reduce noise, followed by the identification of R-peaks using the Pan-tompkins [74]algorithm which has an average about 1%. The data was labeled into three classes, AF, pre-AF and SR.

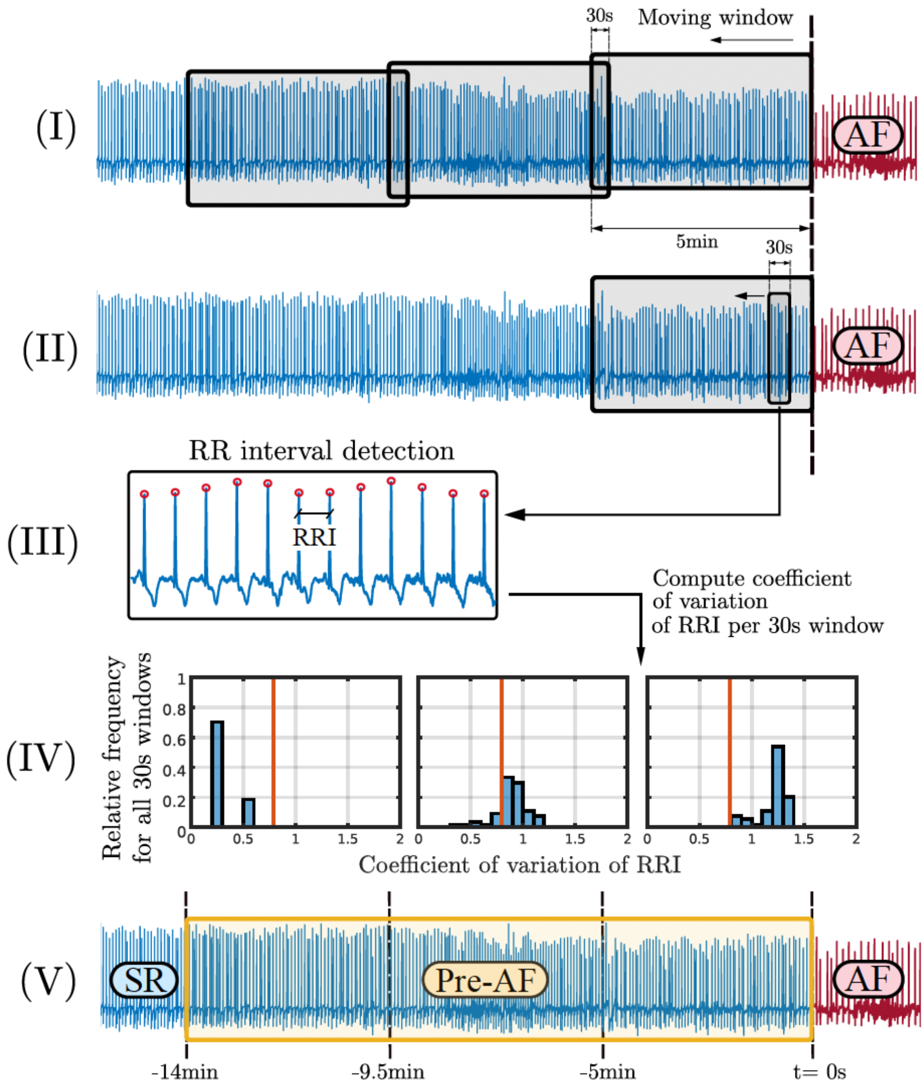

Figure 9: Pre-AF labeling process for a representative patient [73]

The labeling of pre-AF ECG segments was performed in five steps, as illustrated in Figure 9:

1. Starting from the annotated AF onset, a 5-minute sliding window with 30-second overlap was moved backward in time.

2. Within each 5-minute segment, a secondary 30-second sliding window with a 5-second step was applied to extract smaller samples.

3. The Pan–Tompkins algorithm was applied to lead II to detect R waves in each 30-second window, enabling the calculation of R–R intervals (RRI).

4. For each 30-second segment, the coefficient of variation of the RRI was computed, and a histogram was generated for the corresponding 5-minute window.

5. The temporal evolution of the frequency distributions was analyzed until their median dropped below 0.7.

The threshold of 0.7 was selected as the intersection point between the frequency distributions of the coefficient of variation in AF and sinus rhythm (SR) regimes, as derived from the training dataset.

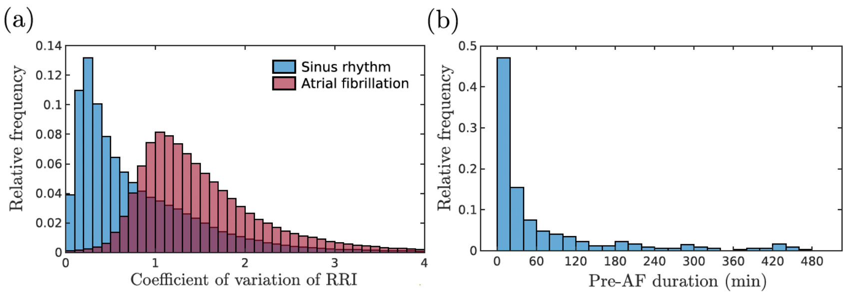

Figure 10: (a) Distributions of the coefficient of variation of the RRI for all patients from the training set, split by SR and AF regimes (as labelled by clinicians). (b) Distribution of Pre-AF length of for patients. [73]

In the last stage, the pre-AF section is segmented, including the beginning of this last window until the onset of AF. Here it mentions that the pre-AF segments vary in length from patient to patient [75] and may also vary within multiple onsets of AF for the same patient due to morphological and electrical changes in the heart over time.

Each 30-second RRI segment was subsequently transformed into a recurrence plot of size 224×224 pixels, which is compatible with convolutional neural network (CNN) architectures commonly employed in image classification tasks. The recurrence plot was generated as the pairwise distance of states in the recorded time-series window, defined as

$$R_{ij} = \|x(i) - x(j)\|, \tag{13}$$

where $\|\cdot\|$ denotes the Euclidean norm, following Takens' theorem with time-delay embedding of the recorded time series [76]. In [73], the embedding dimension (m) and time delay (τ) were set differently for RRI and ECG data: $m = 2$ and $\tau = 3$ for RRI, while $m = 5$ and $\tau = 6$ were used for ECG. These parameter values were chosen based on established criteria, where m corresponds to the first local minimum of the mutual information function [77], and τ represents the smallest value for which the percentage of false nearest neighbors falls below 10% [78].

In Figure 11, panel (a) illustrates recurrence plots corresponding to the three classes: SR, Pre-AF, and AF. When comparing SR and AF, it can be observed that the number of high recurrences decreases on average, as shown in Figure 11 (b). This reduction indicates a weakening of the system's periodicity. In contrast, the Pre-AF state highlights a dynamic transition, characterized by the presence of abundant cross-shaped regions, which arise as a consequence of intermittency during the transition from SR to AF.

The recurrence plots were subsequently used as input to a deep convolutional neural network, EfficientNetV2, a 479-layer architecture developed by Google in 2021 [79, 80]. The network accepts images of size 224×224 pixels, making it well-suited for the recurrence plot representation. The final fully connected layer was modified to enable three-class classification. The model outputs three probabilities—$P(\text{SR})$, $P(\text{Pre-AF})$, and $P(\text{AF})$—representing the likelihood of the input data belonging to each of the respective regimes.

The authors further investigated the impact of varying sample window sizes, ranging from 10 seconds to 5 minutes. The best performance was achieved with a 30-second window, which

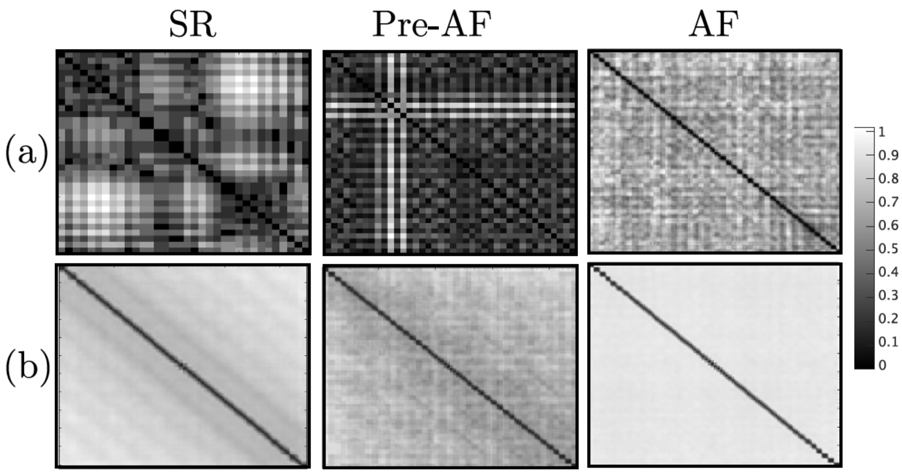

Figure 11: Recurrence plots for different segments (SR, Pre-AF, and AF). (a) Generated from a single 30-second sample. (b) Averaged across all recurrence plots for a representative patient [73]

reflects a trade-off between the number of training samples and the temporal information contained within each window. Longer windows reduce the number of available samples, thereby limiting the ability of WARN to generalize effectively. Conversely, shorter windows may lead to information loss, as they fail to capture sufficient temporal dynamics.

Table 4: Optimal length of the sampling window. [73]

Length (seconds)	Samples ($\times 10^6$)	Accuracy
10	2.3	0.70
30	0.8	0.74
60	0.4	0.72
120	0.2	0.69
300	0.1	0.66

After identifying the optimal window size, the authors compared the performance of WARN with two commonly used architectures for arrhythmia prediction: 1-D CNN and LSTM. The proposed WARN model achieved an average validation accuracy of 0.74, with strong generalization capability reflected by a low standard deviation of 0.03 across all 10 folds.

Table 5: 10-fold cross-validation accuracy after training WARN and two benchmark networks. [73]

Model	1	2	3	4	5	6	7	8	9	10	Average
EfficientNetV2	0.78	0.71	0.73	0.73	0.73	0.71	0.75	0.75	0.76	0.71	0.74
LSTM	0.64	0.61	0.63	0.61	0.61	0.58	0.60	0.63	0.57	0.60	0.60
1D-CNN	0.71	0.68	0.67	0.71	0.69	0.67	0.68	0.67	0.70	0.70	0.69

In the last stage WARN provide an early warning of AF as the probability of a patient switching to AF from the outputs of the trained CNN. The probability of danger is defined as, P(danger) = P(pre-AF)+P(AF), which represents the probability that a sliding window is either in the pre-AF or AF class.

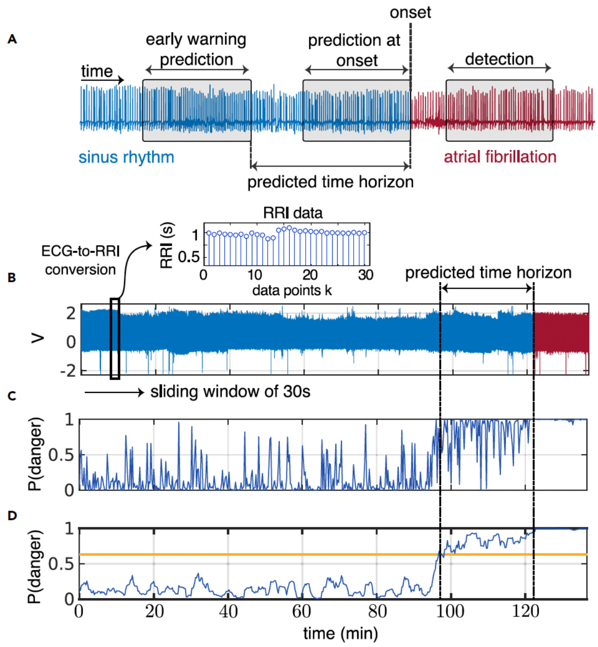

Figure 12: Detection vs. prediction and early warnings. (A) Data windows for early warning, AF onset prediction, and detection; the predicted time horizon is the interval to AF onset. (B) Sliding 30-s ECG windows sampled every 15 s are converted to RRI. (C) The model computes a "probability of danger" for each window. (D) A non-anticipative moving average smooths this probability, triggering a warning when it crosses a threshold; in this example, 24 min before AF onset. [73]

Figure 12(c) illustrates the probability of danger computed by WARN for a representative case. Due to the high variability of this probability, the authors applied a non-anticipative moving average window to filter high-frequency noise and smooth the output, as shown in Figure 12(c–d). A non-anticipative moving average computes the mean of all data points up to and including the current point, without relying on any future values. In addition, the generation of a binary early-warning indicator ("danger" or "normal") requires the selection of a probability threshold.

Thus, two hyperparameters—the moving average window length and the probability threshold—can be optimized to achieve effective early warning performance depending on the needs of particular patients. To optimize

accuracy, F1 score, and mean and median predicted time horizon. The resulting hyperparameters are a moving average window size of 7 samples and a threshold of 0.57.

To optimize the two hyperparameters—the moving average window length and probability threshold—the performance of WARN was evaluated using 60-minute sequences of data before AF onset and during sinus rhythm (SR), selected on a per-patient basis. The 60-minute time horizon was chosen because more than 70% of all pre-AF segments have a duration shorter than one hour. For SR segments, samples were randomly selected at least two hours before AF onset to ensure that the median coefficient of variation of the RRI signals (computed over the selected 60-minute sample) was close to the median value associated with the SR distribution across all patients Figure 13(a).

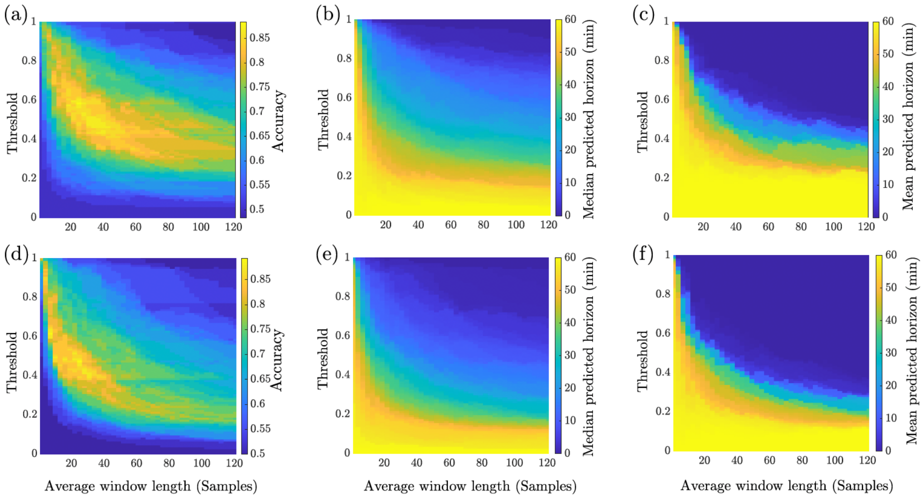

Figure 13: Tradeoffs on the validation data of the WARN model (a) Model accuracy, (b) mean and (c) median of the predicted time horizon before AF onset on the validation set of the RRI data as a function of the probability threshold and the size of the moving average window that smooths the probability of danger. (d, e, f) Same plots computed for the ECG data. [73]

As expected, it was not possible to maximize all performance measures simultaneously. At low probability thresholds, the predicted time horizon until AF onset (the time from the first early warning to AF onset) was maximized Figure 13(c), but this came at the cost of reduced accuracy (bottom of Figure 13(a)). Conversely, when accuracy was maximized (88.3%), the predicted time horizon was relatively short. To achieve a trade-off, the authors searched for hyperparameter settings where accuracy, sensitivity, specificity, and F1 score were all above 80%, while mean and median predicted time horizons exceeded 30 minutes. A total of 34 hyperparameter combinations satisfied this condition.

From these, the smallest moving average window was chosen to minimize computational and memory requirements on smart devices. Among the eligible options, the combination that maximized accuracy, F1 score, and mean/median predicted time horizon was selected. The resulting optimal parameters were a moving average window size of 7 samples (corresponding to 1.5 minutes) and a probability threshold of 0.57 (Section S6). With these settings, WARN achieved an accuracy of 86.7%, F1 score of 87.5%, sensitivity of 93.3%, and specificity of 80%. The mean and median predicted time horizons until AF onset were 31.4 and 36.3 minutes, respectively.

To externally validate WARN, its performance was evaluated on independent ECG datasets collected from patients with AF across three healthcare centers in different countries: Clínica y Maternidad Suizo Argentina (53 patients, 24-hour ECG), Groupe Hospitalier Privé Ambroise Paré – Hartmann in France (250 patients, 24-hour ECG), and the open-access Atrial Fibrillation Prediction Database (AFPDB) from Physionet (75 patients, 30-minute ECG) [67]. A total of 73 patients: 8 from Argentina, 25 from France, and 40 from Physionet was used to externally validate the model.

The Physionet database included 50 healthy controls (SR) and 25 AF patients (30-minute ECG segments immediately before AF onset and 5 minutes after). Among these, 5 AF patients with episodes shorter than 1 minute and 2 healthy-control records affected by AF or severe artifact noise were excluded. Consequently, the external validation dataset consisted of 20 AF records and 20 randomly selected SR records as controls.

Table 6: Model performance on different datasets. [73]

Dataset	Accuracy (%)	AUROC	AUPRC	Predicted time horizon	
				Mean	Median
Test (RRI)	82.7	0.90	0.88	30.8 min	38.0 min
Test (ECG)	82.4	0.95	0.96	32.5 min	43.4 min
External centers	75.0	0.80	0.73	31.8 min	41.3 min

The authors initially validated WARN on a test dataset consisting of RRI data, achieving an overall accuracy of 83% with a probability threshold of 0.57. The model also attained high scores in both the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC).

Analysis of misclassifications revealed that, among the four false negatives, one patient experienced a sudden AF onset following a very stable SR period, while the remaining three patients exhibited a combination of tachycardia, bradycardia, unstable baselines, and noisy signals prior to AF onset. Of the 22 false positives, 13 involved premature atrial contractions (PACs), 5 involved premature ventricular contractions, 6 had unstable baselines, 4 had sinus tachycardia, and 1 exhibited atrial flutter. Notably, the majority of these false-positive records (15 of 22) were highly noisy, highlighting the importance of proper electrode-skin contact, such as using saline or disinfectant, to ensure reliable ECG recordings.

Table 7: Observation on misclassification by WARN in the test dataset. [73]

False Negatives	
Sample	Observation
1	Very stable rhythm, only SR
2	Tachycardia, bradycardia, and noise
3	Tachycardia, unstable baseline and noise
4	Unstable baseline
False Positives	
Sample	Observation
1	Atrial flutter
2, 6, 14, 19	Noise and unstable baseline
3, 8	Multiple PVCs and PACs
4, 15	PACs and noise
5	PACs and sinus tachycardia
7	Sinus tachycardia coupled with PACs and noise
9, 10, 16	Noise
11, 17, 22	PVCs, PACs and noise
12	Sinus tachycardia, multiple PACs and noise
13	Sinus tachycardia and unstable baseline
18	Multiple PACs and noise
20	PACs, unstable baseline and long RR intervals
21	PACs

Beyond the influence of noise, some false-positive events may correspond to moments when the heart was close to transitioning from SR to AF but did not. External factors, such as stress or stimulants, can push cardiac dynamics toward a tipping point that triggers a dynamic transition from SR to AF. This is particularly relevant for patients with PACs (13 of 22 false positives), which are well-known precursors of AF and are strongly associated with its occurrence [81].

The authors also evaluated WARN's performance on the open-access Paroxysmal AF Prediction Challenge Database (AFPDB) from Physionet [67, 82], as summarized. The analysis was conducted on a balanced set of 20 AF patients and 20 healthy controls. It is important to note that WARN was originally trained and tested exclusively on data from patients already diagnosed with AF, which may explain the slightly lower performance on the Physionet dataset—accuracy of 0.7, AUROC of 0.76, and AUPRC of 0.79. Additionally, [73] says that the Physionet dataset contains only 30-minute ECG recordings, resulting in shorter predicted time horizons for AF (mean of 12.9 minutes). Finally, it should be noted that previous models evaluated on Physionet were not always reproducible, complicating direct comparisons [83].

The authors further investigated whether training the CNN directly on the original ECG data would yield substantial performance gains. Overall, using ECG data led to a slight improvement compared to RRI-based training: the AUROC and AUPRC increased by 5.5% and 9.1%, respectively, while accuracy and mean predicted time horizons remained largely similar Table 7.

Given that converting ECG to RRI inherently results in some loss of information, it is expected that training on raw ECG data could improve performance. However, the observed improvement was relatively modest, despite the richer, higher-resolution, continuous-time nature of ECG data compared to the lower-sampled, discrete RRI data. These findings indicate that accurate AF onset prediction can be effectively achieved using only RRI data, highlighting the potential for real-time monitoring and early-warning applications with comfortable wearable devices.

The authors also reported an analysis using PPG data on a cohort of 554 patients, the largest cohort presented in their study. Using 17 engineered features extracted from RRI data, an eXtreme Gradient Boosting (XGBoost) classifier achieved an accuracy of 88% for AF onset prediction.

Table 8: Comparison with previous studies on AF prediction.

Year	Study	Method	Patients (no.)	Window length	Prediction horizon	Accuracy (%)	Sensitivity (%)	Specificity (%)
2012	Mohebbi et al. [84]	RRI, SVM	NR	30 min	onset	96	96	93
2013	Costin et al. [85]	ECG, QRS complexes	75	5 min	onset	90	89	89
2016	Boon et al. [86]	RRI, SVM	53	30 min	onset	80	81	79
2018	Li et al. [87]	ECG, Markov chain	5	2 min	onset	82	86	80
2018	Boon et al. [88]	RRI, SVM	53	5 min	onset	87	86	88
2018	Ebrahimzadeh et al.[39]	ECG, mixture of experts	53	5 min	onset	98	100	96
2021	Guo et al. [89]	RRI, XGBoost	554	1–4 h	onset	88	82	96
2021	Tzoul et al. [90]	ECG, CNN	8	5 min	onset	89	88	89
2022	Grégoire et al. [91]	RRI, CNN	140	300 RRI (∼5 min)	30 beats (∼30 s)	66	80	53
2023	WARN [73]	RRI, CNN	350	30 s	30.8 min before onset	83	95	70
2023		ECG, CNN	350	30 s	32.5 min before onset	82	95	69

The [89] study used PPG to predict AF real time onset among high risk individuals particularly those with paroxysmal AF. The work was based on the Huawei Heart Study [92], a large-scale AF screening initiative employing photoplethysmography (PPG) enabled smart devices. Even though this dataset was used in the study, it is not publicly available, which limits reproducibility and comparative analysis with other methods.

The authors first extracted 17 features from the RRI data, including time-domain, frequency-domain, and non-linear features. Following features were extracted;

Heart Rate Features

- **MinHR**: Minimum value of all RR intervals
- **MeanHR**: Mean value of all RR intervals
- **MedianHR**: Median value of all RR intervals
- **SkRR**: Skewness of all RR intervals

Heart Rate Variability (HRV) Features

- **CVRR**: Coefficient of variation of all RR intervals
- **SDNN**: Standard deviation of all NN intervals
- **SD ratio**: Ratio of standard deviations (not fully detailed in text, but used as HRV metric)
- **pNN50**: NN50 count (number of pairs of successive NNs differing by more than 50 ms) divided by total number of NN intervals

Entropy and Mathematical Features

- **Sample entropy**
- **Poincaré stepping** (nonlinear dynamics feature from Poincaré plot)
- **Frequency domain** features
- **Shannon entropy**
- **Approximate entropy**
- **Difference** (likely variation across intervals)
- **PPG monitoring time**

User-Defined Feature

- **AFprob**: Probability of AF detection output (from the prior Huawei AF detection algorithm, M0)

After comparing multiple classifiers (Random Forest, SVM, Gradient Boosting, XGBoost), these features were then used to train an eXtreme Gradient Boosting (XGBoost) classifier to predict AF onset. The model was trained and validated using a cohort of 554 patients, achieving an AUC of 94% when predicting AF onset 4 hours before AF onset.

To achieve an improved predictive ability in terms of the accuracy, precision, F1 score, and AUC, they have further optimized M1 by incorporating additional features and fine-tuning hyperparameters. These features include,

- **HRR features**
  - Combination of current RR interval features with the **history of RR features** (i.e., incorporating temporal trends rather than only instantaneous values).
- **CHRR features**
  - HRR features combined with **context information**, which provided a broader temporal and situational perspective to capture variations that might precede AF onset.

M2 achieved an AUC of 0.97, accuracy of 0.93, precision of 0.91, F1 score of 0.86, sensitivity of 0.85 and specificity of 0.97 when predicting AF onset 4 hours before AF onset, outperforming M1 and other baseline models.

Table 9: Predictive Ability of the Primary AF ML Model (M1) and the Optimized AF ML Model (M2) [31]

	Test 1		Test 2	
	M1	M2	M1	M2
Accuracy	0.912 (0.910–0.913)	0.935 (0.933–0.936)	0.891 (0.890–0.893)	0.934 (0.933–0.936)
Precision	0.864 (0.860–0.868)	0.914 (0.910–0.917)	0.840 (0.836–0.844)	0.912 (0.909–0.916)
Sensitivity	0.775 (0.772–0.781)	0.821 (0.817–0.825)	0.730 (0.725–0.735)	0.813 (0.809–0.827)
Specificity	0.958 (0.957–0.959)	0.974 (0.973–0.975)	0.950 (0.948–0.951)	0.973 (0.972–0.974)
F1 score	0.818 (0.815–0.821)	0.865 (0.862–0.868)	0.781 (0.778–0.784)	0.865 (0.863–0.868)
AUC[a]	0.942 (0.933–0.943)	0.971 (0.970–0.981)	0.932 (0.923–0.935)	0.972 (0.971–0.982)

Furthermore, M2 reduced false-positive rates compared to M1, though many false alarms were attributable to atrial arrhythmias other than AF, such as bigeminy and flutter. This finding highlights the challenge of distinguishing AF onset from other atrial rhythm disturbances.

However, several limitations remain. First, the model detects pre-AF signals from sinus rhythm, but the proprietary nature of the dataset, limited details about the model architecture, and the lack of information on deployment or testing on edge devices raise concerns about transparency and reproducibility. Second, validation was carried out only in relatively small, high-risk cohorts, leaving uncertainty about its generalizability to broader populations. Finally, although predictive accuracy was high, the limited explanation of why and how specific features were extracted, and how they contributed to the model's predictions, remains unclear.

4 Research Scope and Objectives

4.1 Project Motivation

AF is the most common arrhythmia in the aging populations where it's paroxysmal and asymptomatic nature makes AF difficult to diagnose in the early stages. Having AF over time increases the risk of strokes, heart failures, congitive decline and overall mortality as well. Therefore the healthcare costs and the clinical burden of AF is profound. Timely detection and prediction of AF can help patients to manage AF with anti-cogulant medicine and heart therapies which allow patients to mitigate the adverse outcomes of AF.

The critical challenge lies not merely in detecting AF after onset, but in predicting its occurrence during normal sinus rhythm, thereby enabling preventive measures before severe complications arise. While traditional risk prediction models and clinical scoring methods are widely used to predict AF, they often demonstrate only modest predictive performance and are not sufficient to make proactive intervention in high-risk individuals. ML models trained on continuous PPG or ECG signals hold promise in addressing this gap, as they can uncover subtle signal patterns that precede AF episodes and outperform conventional approaches.

Recent advances in wearable technologies, such as PPG-based smart devices and continuous monitoring patches, have provided new opportunities for large scale AF detection and screening. These devices allow the capture of high-resolution physiological signals in real-world environments, beyond the limited scope of hospital based monitoring. Predictive systems deployed at edge allow processing AF related signals directly on the wearable or on the device which provides significant advantages; (1) enables real-time analysis with minimal latency, (2) ensures data privacy by reducing dependence on cloud storage, and (3) reduces energy consumption and costs associated with continuous data transmission. Moreover, edge based prediction (4) ensures robustness in environments with unreliable network connectivity, expanding the accessibility of AF monitoring to remote or resource constrained settings.

By integration of advanced ML algorithms with edge computing, continuous and personalized AF risk assessment becomes feasible, empowering patients with timely alerts and enabling clinicians to intervene early. This motivated us to build a solution which is cost-effective and preserves privacy in AF management.

4.2 Problem Statement

Despite advances in wearable technology and machine learning, current AF detection systems primarily identify AF after onset and rely heavily on cloud-based processing, raising concerns about latency, privacy, and accessibility. To the best of our knowledge, there is a lack of reliable, real-time, edge-deployable predictive models capable of anticipating AF episodes during normal sinus rhythm, particularly in resource-constrained or low-connectivity environments. Moreover, existing approaches do not provide insights into how and why predictions are made, nor do they quantify the contribution of individual features to the model's output, limiting their interpretability, replicability and trustworthiness. This gap highlights the urgent need for robust, efficient, privacy-preserving, and explainable AF prediction solutions that can operate directly on wearable devices.

4.3 Research Objectives

Building upon the identified gaps in current AF prediction systems, this research focuses on designing an edge-deployable, real-time solution that not only anticipates AF onset but also provides transparency in its predictions. By leveraging wearable sensor data and machine learning techniques, the study aims to address limitations in latency, energy efficiency, connectivity, and explainability, thereby enabling timely and personalized AF management. The following specific research objectives guide this work:

- Develop a real-time, edge-deployable model to predict AF episodes during normal sinus rhythm using PPG and ECG data.
- Optimize the model for resource-constrained wearable devices to ensure low latency and minimal energy consumption.
- Integrate explainable AI techniques to provide transparency, clarifying how predictions are made and the contribution of individual features.
- Validate the model on real-world datasets to assess predictive performance, including accuracy, sensitivity, specificity, and early warning time horizon.
- Ensure privacy-preserving operation by minimizing reliance on cloud-based processing.

4.4 Research Questions

Based on the identified problem and the objectives of this study, the following research questions guide this work:

1. Can atrial fibrillation (AF) onset be accurately predicted in real-time using wearable sensor PPG data?
2. How can machine learning models be optimized and compressed to perform efficiently on resource-constrained edge devices while maintaining predictive accuracy?
3. Which physiological, mathematical, and statistical features contribute most to AF prediction, and how can explainable AI techniques clarify their role in the model's decision-making?
4. Can an edge-based predictive system achieve comparable performance to cloud-based solutions while preserving user privacy and minimizing latency?
5. What is the optimal pipeline, including data processing, feature extraction, model deployment, and inference, for running a predictive AF algorithm on resource-constrained devices?

5 Methodology

5.1 Approach

Our approach focuses on predicting atrial fibrillation efficiently at the edge using wearable devices equipped with PPG sensors. The system is designed as a hierarchical pipeline with two sub-pipelines that balance lightweight continuous monitoring with more computationally intensive but accurate analysis, triggered only when abnormalities are detected.

5.1.1 Data Acquisition

Data acquisition is performed using the onboard PPG sensor of the wearable device (e.g., fitness tracker or smartwatch). A lightweight application running on the wearable continuously collects raw PPG data, ensuring low-latency streaming and minimal energy overhead.

Datasets

PAF Prediction Challenge Database (AFPDB)

The **Paroxysmal Atrial Fibrillation Prediction Challenge Database (AFPDB) [67]** is a collection of two-channel ECG recordings created for the Computers in Cardiology Challenge 2001, an open competition aimed at developing automated methods for predicting paroxysmal atrial fibrillation (PAF). The dataset is organized into three main groups:

1. **Non-AF group**: This consists of 100 ECG recordings of healthy individuals, each 30 minutes in duration, labeled *n01–n50* and *n01c–n50c*. These records contain no atrial fibrillation events.

2. **Pre-AF group**: This includes ECG recordings from 25 patients diagnosed with PAF. Each patient record set contains two 30-minute segments obtained prior to AF onset. The first segment (*p01, p03 . . . p49*) was recorded within 30 minutes before the onset of AF, while the second segment (*p02, p04 . . . p50*) was recorded at least 45 minutes before AF onset. These are followed by corresponding 5-minute AF episode segments.

3. **AF group**: This contains the 5-minute ECG segments (*p01c, p02c . . . p49c, p50c*) that capture the actual AF episodes immediately following the Pre-AF segments.

All ECG signals were originally sampled at 128 Hz with a resolution of 12 bits. The temporal division of records into Non-AF, Pre-AF, and AF groups enables the analysis of physiological variations outside, prior to, and during AF events.

For this study, 100 Non-AF records and 50 Pre-AF records were utilized. To avoid patient overlap across subsets, these records were partitioned into training, validation, and test sets in a ratio of 7:2:1. The signals were then resampled to 500 Hz and segmented into 10-second windows, yielding 1575 samples for training, 450 samples for validation, and 225 samples for testing.

MIMIC PERform AF Dataset

The MIMIC PERform AF dataset is a dataset that can be used for developing and evaluating atrial fibrillation (AF) detection algorithms using photoplethysmogram (PPG) signals. PPG, commonly measured by pulse oximeters and wearable devices such as smartwatches. This dataset contains annotated PPG recordings with corresponding electrocardiogram (ECG) references, enabling reliable benchmarking of beat detection and AF classification algorithms. Furthermore, it includes recordings from patients in both normal sinus rhythm and AF, allowing algorithms to learn and distinguish irregular heartbeat patterns. The dataset supports evaluation under various physiological conditions and patient demographics, providing a robust foundation for real-world AF detection applications.

5.1.2 Model Training and Optimization

Transfer learning and model compression.

The first strategy involves leveraging transfer learning by adapting the pre-trained WARN model, originally developed on ECG-derived R-R interval (RRI) data, to photoplethysmography (PPG) peak-to-peak interval signals. This approach enables the transfer of temporal and physiological representations learned from ECG to PPG, which is particularly advantageous given their shared cardiovascular dynamics. To further adapt WARN for edge deployment, knowledge distillation will be employed in a teacher–student paradigm, where the larger WARN model serves as a teacher to guide a lightweight student network. Subsequent model compression and optimization techniques (e.g., pruning, quantization) will be applied to ensure that the student model can efficiently operate on resource-constrained devices.

Classical machine learning models.

As an alternative strategy, models widely reported in the literature—such as XGBoost, decision trees, and support vector machines (SVM)—will be trained directly on the found PPG datasets, thereby replicating the existing work. By analysing which features contribute most to the model's predictions, we can establish a transparent mechanism for feature selection, allowing the identification of the most predictive features contributing to atrial fibrillation (AF) prediction.

Performance comparison.

The performance of both approaches will be systematically compared in the context of edge deployment. This evaluation can be conducted either by deploying the models on resource-limited devices or by simulating such environments using control groups (cgroups). The comparative analysis will highlight trade-offs between accuracy, computational efficiency, and interpretability, ultimately guiding the selection of the most suitable model architecture for AF detection on edge devices.

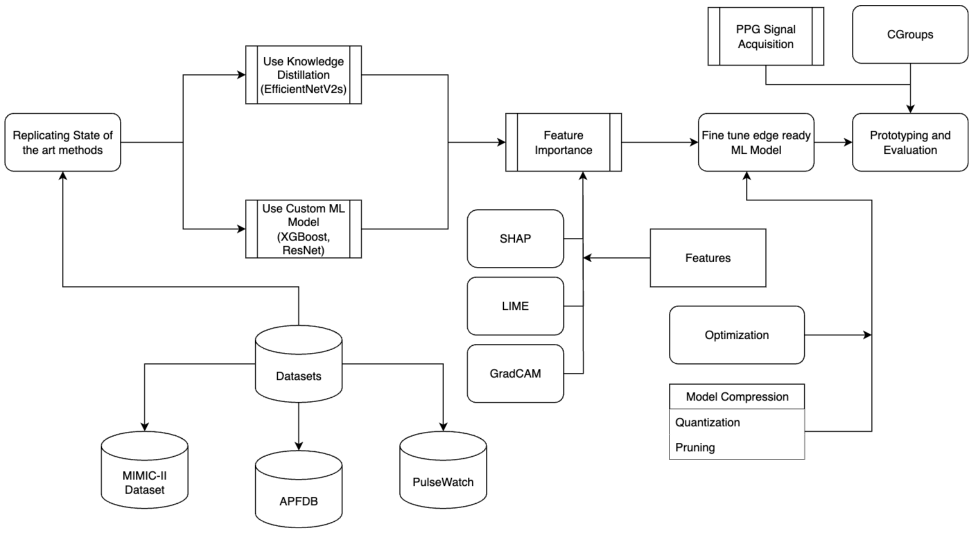

Figure 14: Overview of the model training and optimization strategies for PPG-based AF detection, including transfer learning with WARN and classical machine learning approaches.

5.1.3 Data Preprocessing

The raw PPG is often contaminated with baseline wander, motion artifacts, and hypoperfusion, we apply a multi-stage pre-processing pipeline to clean the signal before any analysis. First, a finite impulse response (FIR) bandpass filter is used to suppress baseline drift and high-frequency noise. Next, empirical mode decomposition (EMD) or wavelet decomposition is applied to further reduce motion and perfusion artifacts while retaining physiologically relevant dynamics. To ensure that only reliable signals are used, we incorporate a signal quality index (SQI) based on skewness and kurtosis, which rejects segments that are heavily distorted. Cleaned and validated signals are then buffered for downstream processing.

5.1.4 Feature Extraction

Feature extraction is carried out in two layers to accommodate both efficiency and accuracy. To ensure the system's suitability for edge deployment, we focus on computationally efficient features that can be calculated in real-time on resource-constrained devices.

Feature Importance and Selection

To assess the contribution of each feature to the model's predictions we compute SHAP (SHapley Additive exPlanations) values. Features with low SHAP values across multiple samples are considered less important and can be pruned from the feature set. Additionally, we perform correlation analysis to identify and remove redundant features that do not provide unique information. By combining these techniques, we can reduce the feature set to a more manageable size without significantly impacting model performance. We hope this combined approach reduces the feature set to a more compact and efficient representation without significantly compromising model performance.

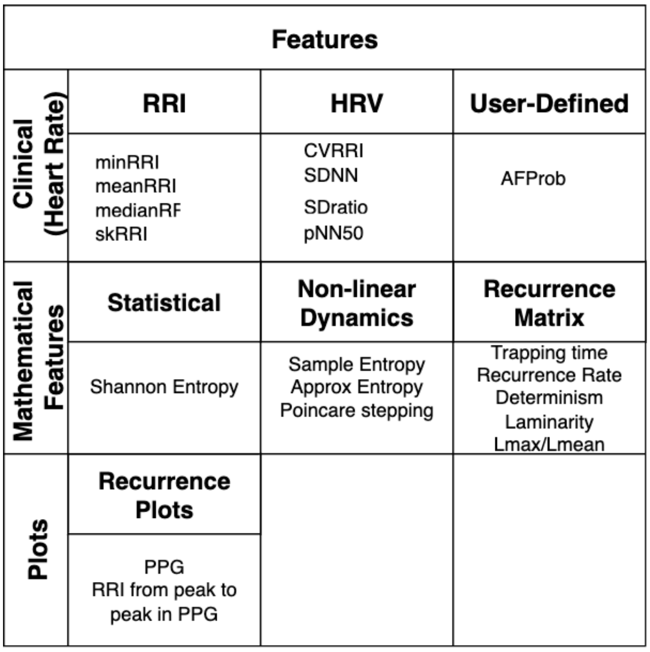

Figure 15: Features analysed in this study.

5.1.5 Pipeline

From the features identified in the previous step, we first select those that contribute most to the model's predictions. Among these, we then determine which features show the greatest deviation between pre-AF windows and SR (normal) windows. The most deviating features are chosen for the first pipeline. The first pipeline, dedicated to continuous screening, relies on threshold-based mechanisms or simple decision trees applied to the lightweight features. Its primary goal is to detect deviations in inter-beat intervals that may indicate premature atrial contractions (PACs) or other irregular patterns that could precede atrial fibrillation. Once such anomalies are detected, the second pipeline is activated. The second pipeline integrates a compact machine learning model, such as a random forest, support vector machine, or a TinyML-optimized neural network, which leverages the extended feature set to more reliably identify early AF conditions. To minimize false positives, the model requires repeated confirmations across multiple inference windows before alerting the user. If the abnormality persists and the condition is deemed clinically relevant, the system notifies the user on-device and optionally logs the event for clinical review. Furthermore, the model performance can be validated using AF detection models from prior studies [28–30]. Note that our focus in not on the detection algorithm, but rather improving the above mentioned prediction pipeline.

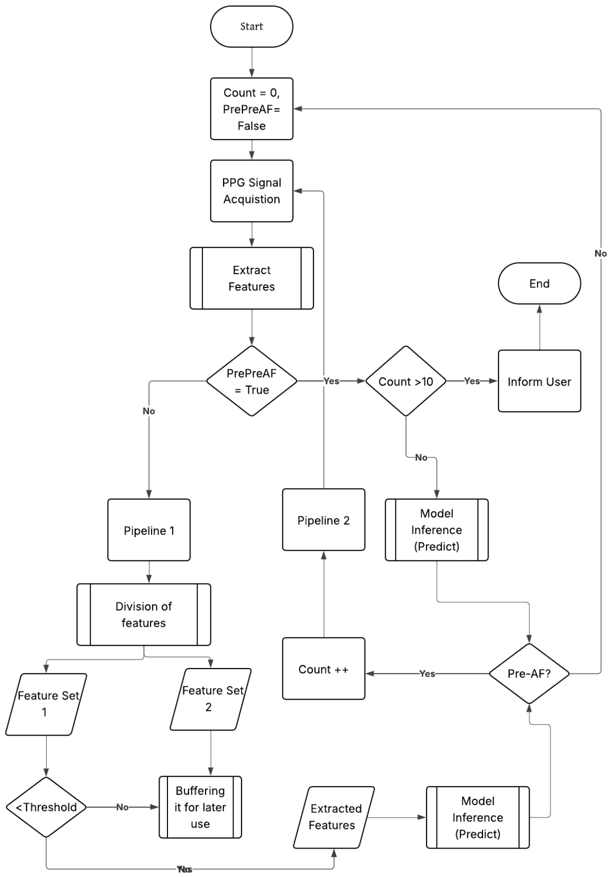

Figure 16: Proposed hierarchical pipeline for AF prediction at the edge.

5.2 Project Phases and Milestones

The methodology implementation is structured into four phases, reflecting the hierarchical pipeline approach for PPG-based AF prediction and optimization for edge devices:

5.2.1 Phase 1: Research and Baseline Establishment

- Conduct an in-depth review of state-of-the-art AF prediction models, including WARN and classical ML approaches.
- Replicate key existing models using the AFPDB and MIMIC PERform AF datasets to establish baseline performance.
- Evaluate different models to identify best practices for feature extraction, preprocessing, and pipeline design.

Milestone 1 Deliverable: Comprehensive benchmark of existing models and a validated baseline pipeline.

5.2.2 Phase 2: Data Preprocessing and Feature Engineering

- Implement multi-stage preprocessing for raw PPG signals, including FIR bandpass filtering, empirical mode decomposition or wavelet filtering, and signal quality indexing.
- Extract computationally efficient features suitable for real-time edge deployment.
- Apply SHAP analysis and correlation-based pruning to reduce the feature set while retaining predictive performance.

Milestone 2 Deliverable: Cleaned, validated PPG data and a compact feature set ready for pipeline integration.

5.2.3 Phase 3: Hierarchical Pipeline Design and Model Optimization

- Design a two-level pipeline:
  - **First Pipeline:** Continuous monitoring using lightweight features and threshold-based or simple decision-tree models.
  - **Second Pipeline:** Activated on anomaly detection, leveraging a compact ML model (e.g., TinyML neural network, random forest, or SVM) with the extended feature set.
- Optimize models for edge deployment using transfer learning, knowledge distillation, pruning, and quantization.
- Conduct simulation or control-group testing to evaluate memory and computational constraints on constrained hardware.
- Generate comparative performance plots for baseline vs. optimized models.

Milestone 3 Deliverable: Optimized hierarchical pipeline tested in simulated edge conditions with validated predictive performance.

5.2.4 Phase 4: Edge Deployment and Real-Time Evaluation

- Deploy the optimized pipeline onto wearable devices with PPG sensors.
- Ensure real-time inference while maintaining computational and memory efficiency.
- Conduct extensive testing using real-world PPG data, comparing results with prior studies and baseline models.
- Monitor system for false positives, repeated confirmations, and user notifications for clinically relevant events.

Milestone 4 Deliverable: Fully deployed real-time AF prediction pipeline on wearable hardware with validated accuracy, efficiency, and robustness.

6 Research Timeline

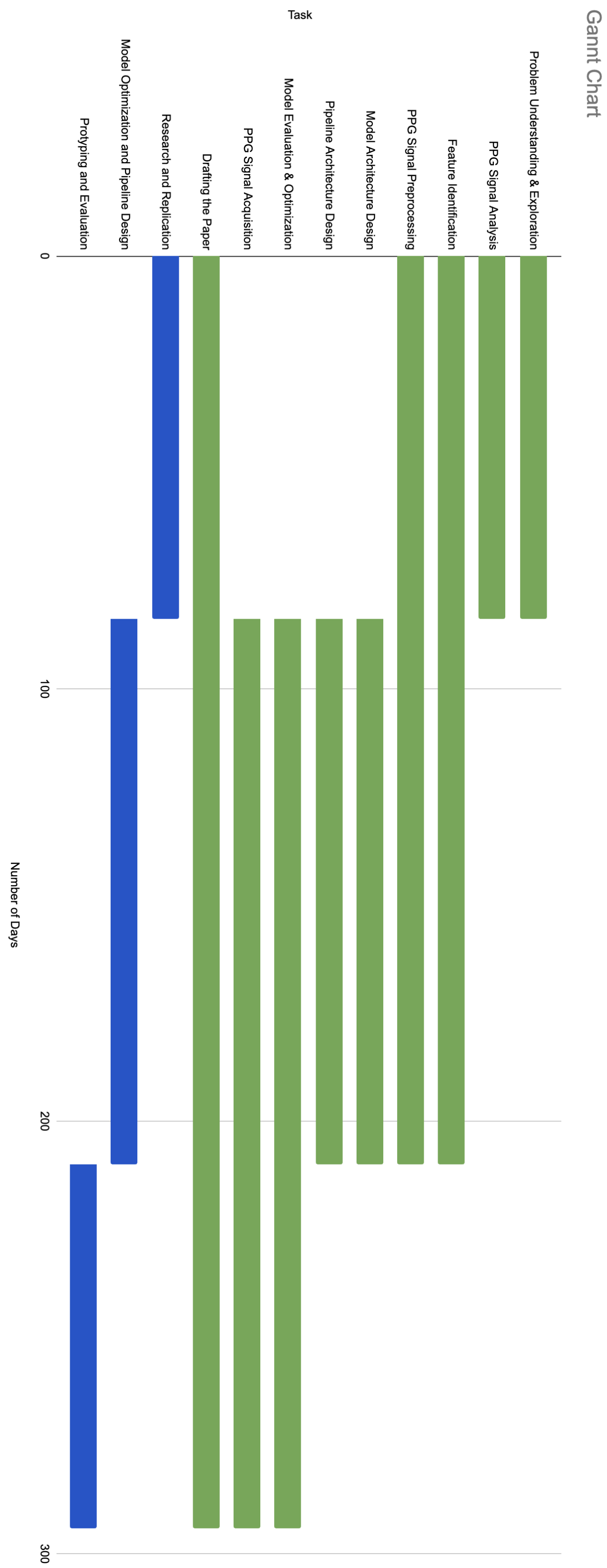

Figure 17: Timeline

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